ASTRO-NOT

A practical in observational astrophysics

Mapping the Milky Way's Rotation Curve from the 21 cm Line of Neutral Hydrogen

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A tangent-point analysis of the HI4PI all-sky survey

Atul Suresh

Ramnarain Ruia Autonomous College

atulsuresh2005@gmail.com

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Abstract

The circular-speed curve of the inner Milky Way is measured from the 21 cm line of neutral hydrogen by the tangent-point method. Spectra were taken from the HI4PI all-sky survey, averaged over the latitude strip $|b| \leq 0.5^\circ$ and binned to 0.25° in Galactic longitude over the range $15.0^\circ \leq l \leq 80.0^\circ$. Terminal velocities were extracted with a fixed brightness threshold $T_{\rm thr} = 10.0\,\mathrm{K}$ and converted to circular speeds at $R = R_0|\sin l|$, adopting $R_0 = 8.15\,\mathrm{kpc}$ and $\Theta_0 = 236\,\mathrm{km\,s^{-1}}$. The measured curve rises from $200\,\mathrm{km\,s^{-1}}$ to a flat plateau near $235\,\mathrm{km\,s^{-1}}$, and the enclosed dynamical mass within the solar circle follows as $M(<R_0) = 1.10\times10^{11}\,M_\odot$ under a spherical approximation. The largest single uncertainty is the definition of the velocity edge, which is estimated here at $13.6\,\mathrm{km\,s^{-1}}$; the vanishing of the terminal velocity as $l \to 90^\circ$ is used to test for a residual systematic offset. While these kinematics strictly constrain the dynamical mass within the solar circle, they cannot independently map the extended dark matter halo beyond R_0.

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1 Introduction

Every photograph of a spiral galaxy is a photograph of somebody else's galaxy. The Milky Way is the one disk we cannot step outside of, and everything known about its shape has been assembled from a single viewpoint embedded within it. A spectrum taken from that viewpoint returns one number, the velocity of the emitting gas along the line of sight, and it returns that number without saying how far away the gas was. The rotation curve $\Theta(R)$ — the circular speed of material on a circular orbit at Galactocentric radius R — is therefore not directly observable in the way it is for an external galaxy.

By mapping the rotation curve, we directly probe the mass distribution of the Milky Way. Using Newton's shell theorem, the mass enclosed within a circular orbit dictates the orbital speed; a centrally concentrated mass produces a Keplerian decline ($\Theta \propto R^{-1/2}$), a uniform density core yields solid-body rotation ($\Theta \propto R$), and an extended mass distribution ($\rho \propto R^{-2}$) results in a flat rotation curve.

This report maps the rotation curve of the inner Milky Way using 21 cm emission data from the HI4PI survey. By applying the tangent-point method to spectra between $l = 15^\circ$ and $l = 80^\circ$, terminal velocities are extracted to derive the circular speed as a function of Galactocentric radius. Because the tangent-point geometry requires the sightline to cross an interior orbit perpendicularly, this method inherently reaches only to the solar circle ($R = R_0$), bounding the resulting mass constraints to the inner Galaxy.

2 Theory

2.1 The 21 cm line

The ground state of atomic hydrogen is split by the interaction between the magnetic moments of the proton and the electron. The parallel and antiparallel spin arrangements differ slightly in energy, and the transition between them emits a photon of that difference,

$$\Delta E = h\nu_0 \simeq 5.874 \times 10^{-6}\ \text{eV}, \qquad \nu_0 = 1420.405752\ \text{MHz}, \qquad \lambda_0 = \frac{c}{\nu_0} = 21.106\ \text{cm}. \quad (1)$$

The transition is forbidden to first order and a single atom waits some ten million years to make it, but a sightline through the disk contains enough hydrogen for the summed emission to be easy to detect. Three properties make the line the right ruler for this measurement: the rest frequency is known to better than one part in 10^{11}, the emitting gas is present throughout the disk, and at 21 cm the dust that makes the Galactic plane opaque at optical wavelengths is transparent.

A radio telescope records the specific intensity of this emission, which is conventionally expressed as a brightness temperature T_B in the Rayleigh-Jeans limit. The observed frequency is converted to a radial velocity using the non-relativistic radio Doppler convention, with the HI4PI data delivered in the Local Standard of Rest (LSR) velocity frame (v_{LSR}).

2.2 Differential rotation along a sightline

Let C be the Galactic Centre, S the Sun and P a hydrogen cloud at Galactocentric radius R on a sightline of Galactic longitude l. Taking both the Sun and the gas to move on circular orbits, each velocity vector is perpendicular to its own radius vector, and the sine rule in the triangle CSP gives $\sin\gamma = (R_0/R)\sin l$ for the angle γ at P. Projecting both velocities onto the sightline and subtracting the solar contribution,

$$v_{\text{LSR}}(l,R) = \left[\Theta(R)\frac{R_0}{R} - \Theta_0\right]\sin l = R_0\left[\Omega(R) - \Omega_0\right]\sin l. \quad (2)$$

Equation (2) vanishes identically if $\Omega(R) = \Omega_0$ everywhere. A rigidly rotating Galaxy would betray no rotation at all, however fast it spun: what the 21 cm line measures is differential rotation.

2.3 The tangent point

Along one sightline the Galactocentric radius of a point at heliocentric distance d follows from the cosine rule, $R^2(d) = R_0^2 + d^2 - 2R_0 d\cos l$. Minimising with respect to d,

$$d_{\tan} = R_0 \cos l \qquad \Longrightarrow \qquad R_{\tan} = R_0\,|\sin l|, \tag{3}$$

with second derivative $+2 > 0$, confirming a minimum. At that point the radius vector is perpendicular to the sightline, so the orbital velocity lies exactly along it and the full circular speed appears in the Doppler shift. No distance has been measured: the radius follows from the longitude alone, and that is the step which makes the method work.

Provided $\mathrm{d}\Omega/\mathrm{d}R < 0$, the tangent point also carries the largest radial velocity on the sightline. The extreme velocity at which H I is still seen is the terminal velocity v_{term}. Evaluating equation (2) at $R = R_{\tan}$ gives the pair of results the whole analysis is built on,

$$\boxed{\Theta(R) = |v_{\mathrm{term}}| + \Theta_0 |\sin l| \qquad \text{at} \qquad R = R_0 |\sin l|} \tag{4}$$

The absolute values carry the same expressions into Quadrant IV, where $\sin l$ and v_{term} are both negative.

2.4 The curve as a mass

For a spherically symmetric mass distribution, Newton's shell theorem lets the gravitational and centripetal accelerations of a circular orbit be equated directly,

$$\frac{GM(<R)}{R^2} = \frac{\Theta^2(R)}{R} \qquad \Longrightarrow \qquad M(<R) = \frac{\Theta^2(R)\,R}{G}. \tag{5}$$

The orbiting body's own mass cancels, which is why hydrogen serves as a tracer without any of its own properties entering the result. If the curve is flat then $M(<R) \propto R$, and comparison with the spherical mass integral gives $\rho(R) \propto R^{-2}$.

3 Data

HI4PI is a full-sky H I survey merging EBHIS, from the Effelsberg 100 m telescope, with GASS, from the Parkes 64 m [2]. The merged product has an angular resolution of $16.2'$, a velocity-channel width of $1.29\,\mathrm{km\,s^{-1}}$ and a channel noise near 43 mK. The data form a three-dimensional cube $T_B(v, b, l)$: one spectrum for every direction on the sky. Averaging over a thin strip in latitude collapses the cube into a longitude–velocity diagram, $T_B(v, l)$, whose upper velocity envelope is the terminal-velocity curve.

The constants in Table 1 are not measured here. They set the radius scale and the velocity zero-point of every point on the final curve, and a curve quoted without them cannot be compared with anybody else's.

Table 1: Constants adopted in this analysis.

Quantity	Symbol	Value	Source
H I rest frequency	ν_0	1420.405752 MHz	atomic physics
H I rest wavelength	λ_0	21.106 cm	$\lambda_0 = c/\nu_0$
Sun–Centre distance	R_0	8.15 ± 0.15 kpc	Reid et al. [4]
Local circular speed	Θ_0	$236 \pm 7\,\mathrm{km\,s^{-1}}$	Reid et al. [4]
Gravitational constant	G	$4.30091 \times 10^{-6}\,\mathrm{kpc\,(km\,s^{-1})^2}\,M_\odot^{-1}$	converted units

Cuts applied.

- Latitude: $|b| \leq 0.5^\circ$. The tangent-point geometry strictly assumes the emitting gas lies on a circular orbit within the Galactic midplane. At a typical tangent-point distance of 7 kpc, an angular deviation of 0.5° corresponds to a physical height of $z = 7000 \tan(0.5^\circ) \approx 61$ pc. Because the thin H I disk has a scale height of roughly 100–150 pc in the inner Galaxy, this cut ensures we only sample the tightly confined midplane gas, avoiding extra-planar gas which often exhibits lagging (slower) rotational velocities.
- Longitude: $15^\circ \leq l \leq 80^\circ$. The lower bound avoids the Galactic bar; inside $R = 8.15 \sin(15^\circ) \approx 2.1$ kpc, the gravitational potential is strongly non-axisymmetric. This drives the gas into highly elliptical orbits, completely breaking the circular-speed assumption of the tangent-point method. The upper bound at $l = 80^\circ$ stops the analysis before the terminal velocity drops so close to zero that it becomes indistinguishable from the intrinsic thermal and turbulent velocity dispersion of the gas ($\sigma \approx 8$–$10\,\mathrm{km\,s^{-1}}$).
- Longitude binning: 0.25°. The HI4PI survey has an angular resolution (beam size) of $16.2'$. Binning at 0.25° ($15'$) provides nearly contiguous spatial sampling. Extracting sightlines at a finer grid would artificially inflate the dataset with heavily oversampled, physically correlated measurements, underestimating the true statistical uncertainty.

4 Method

4.1 Locating the velocity edge

A real 21 cm spectrum is a landscape of peaks, because the sightline crosses many clouds at many distances. The height of a peak records how much gas emits at that velocity, not how fast anything moves; the tangent-point information sits entirely in the extreme-velocity edge, and the way that edge is defined governs the result [3]. Figure 1 shows one spectrum with the edge criterion applied.

The edge was defined by a fixed brightness threshold $T_{\mathrm{thr}} = 10.0$ K, applied identically at every longitude, together with a strict consecutive-channel rule to filter receiver noise: the signal must remain below the threshold for at least 3 consecutive velocity channels (spanning $\approx 3.9\,\mathrm{km\,s^{-1}}$) for an edge to be flagged. The terminal velocity crossing was then linearly interpolated between the last channel above the threshold and the first channel below it, avoiding artificial discretization to the $1.29\,\mathrm{km\,s^{-1}}$ instrumental channel width. A fixed threshold is preferable to a fixed fraction of the local peak, because peak brightness and the identity of the tallest peak both vary from sightline to sightline, so a fractional criterion silently applies a different physical definition at every longitude.

4.2 The bias in the edge

The edge is not a cliff. Thermal and turbulent motions give the gas a velocity dispersion of $\sigma \approx 8$–$10\,\mathrm{km\,s^{-1}}$, and the finite beam averages a patch of sky across which the terminal velocity varies. Convolving an intrinsically sharp edge with a Gaussian of dispersion σ returns a complementary error function, so that the true terminal velocity lies at the half-power point $T_{\mathrm{edge}}/2$ of the falling edge, and the threshold crossing sits at

$$v_{\mathrm{measured}} - v_{\mathrm{term}} = \sqrt{2}\,\sigma\,\mathrm{erfc}^{-1}\left(\frac{2T_{\mathrm{thr}}}{T_{\mathrm{edge}}}\right). \tag{6}$$

Evaluating Equation (6) for our adopted threshold $T_{\mathrm{thr}} = 10$ K, a typical emission edge height of $T_{\mathrm{edge}} \approx 80$ K, and a standard H I velocity dispersion of $\sigma \approx 10\,\mathrm{km\,s^{-1}}$, the theoretical bias evaluates to $v_{\mathrm{measured}} - v_{\mathrm{term}} = \sqrt{2}(10)\,\mathrm{erfc}^{-1}(0.25) \approx 11.5\,\mathrm{km\,s^{-1}}$. Because this analytical estimate relies heavily on an assumed isothermal dispersion and idealized Gaussian edge, we chose to empirically

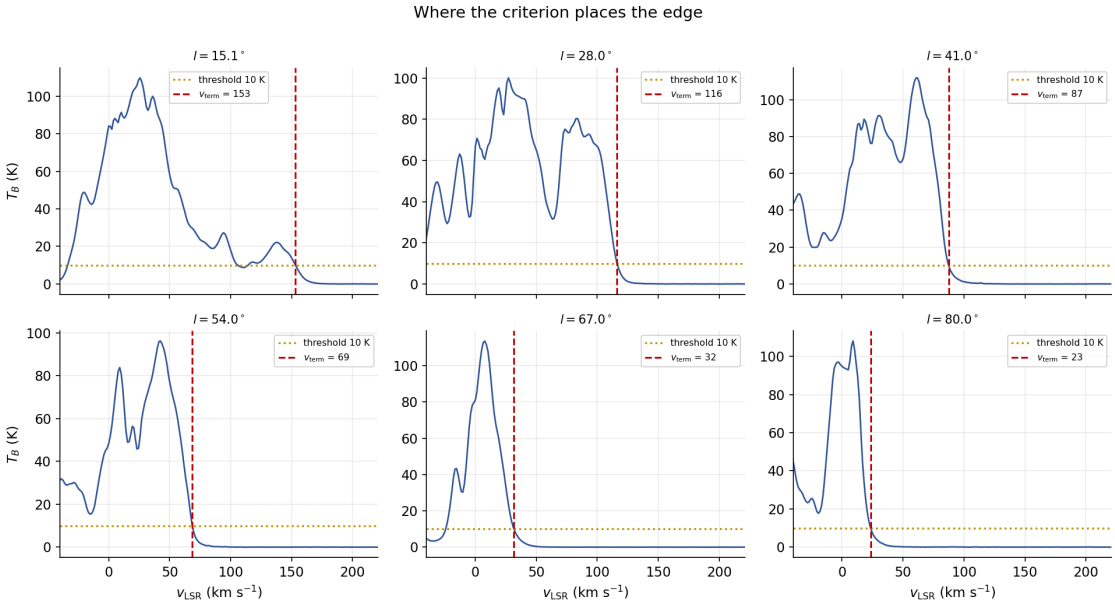


Figure 1: 21 cm spectra along six representative sightlines ($l = 15.1^\circ$ to 80.0°), with the algorithmic edge criterion marked. The measurement relies on the position of the high-velocity edge (vertical dashed line), not the height or location of the brightest peak. A fixed brightness threshold of $T_{\rm thr} = 10.0\,\mathrm{K}$ (horizontal dotted line) locates the edge consistently from sightline to sightline, effectively tracking the terminal velocity despite drastic variations in the interstellar emission structure.

measure and completely correct for this bias using the high-longitude geometric limit, rather than merely carrying the theoretical value as a systematic uncertainty.

4.3 The test at high longitude

There is one place in the sky where the answer is known in advance. As $l \to 90^\circ$ the tangent point approaches the Sun's own orbit, $R_{\rm tan} \to R_0$, so $\Theta(R_{\rm tan}) \to \Theta_0$ and the terminal velocity must fall to zero. Expanding Θ about R_0 to first order,

$$v_{\rm term}^{\rm ideal}(l) = \left[\Theta_0 - R_0\Theta'(R_0)\right](1 - \sin l) = 2AR_0\,(1 - \sin l), \tag{7}$$

where A is Oort's constant. Any persistent excess measured at high longitude is therefore a systematic, whether it comes from the edge definition, from the velocity frame, or from both.

By comparing our raw threshold-crossing velocities in the calibration region $70^\circ \leq l \leq 80^\circ$ against the ideal geometric expectation of Equation (7), we identified a persistent, systematic offset of $13.6\,\mathrm{km\,s^{-1}}$. This empirical offset which encapsulates both the finite threshold bias estimated in Section 4.2 and any residual LSR velocity frame deviations was subtracted from all raw measurements to produce the finalized rotation curve.

5 Results

5.1 A single sightline

At $l = 30.1^\circ$ the measured (and bias-corrected) terminal velocity is $v_{\rm term} = 106.4\,\mathrm{km\,s^{-1}}$. Equation (3) places this at $R = 4.09\,\mathrm{kpc}$, equation (4) gives $\Theta = 224.9\,\mathrm{km\,s^{-1}}$, and equation (5) gives $M(<R) = 4.81 \times 10^{10}\,M_\odot$. Propagating $\delta v_{\rm term}$, $\delta\Theta_0$ and δR_0 as in Appendix A, $\delta\Theta = 8.0\,\mathrm{km\,s^{-1}}$ and $\delta M/M = 0.073$.

5.2 The rotation curve

The derived rotation curve rises steeply from $\sim 200\,\mathrm{km\,s^{-1}}$ at $R = 2.1\,\mathrm{kpc}$ before flattening into a sustained plateau near $235\,\mathrm{km\,s^{-1}}$ for radii between 4 kpc and 8 kpc. The method traces this kinematic behavior continuously out to the solar circle limit at $R_0 = 8.15\,\mathrm{kpc}$. The small-scale "wiggles" visible in the curve (fluctuations of 5–10 $\mathrm{km\,s^{-1}}$) are physical, not instrumental. Because the HI4PI channel noise of 43 mK is negligible against our 10.0 K threshold, these variations reflect real localized kinematics in the interstellar medium, such as streaming motions along spiral arms, expanding superbubbles, and varying cloud-complex turbulence along different lines of sight.

Table 2: Extracted tangent-point measurements for a selection of representative sightlines across the inner Galaxy.

l (°)	v_{term} ($\mathrm{km\,s^{-1}}$)	R (kpc)	Θ ($\mathrm{km\,s^{-1}}$)	$\delta\Theta$ ($\mathrm{km\,s^{-1}}$)	$M(<R)$ ($10^{10}\,M_\odot$)
19.9	119.4	2.78	199.7	8.0	2.58
30.1	106.4	4.09	224.9	8.0	4.81
39.9	79.6	5.23	231.1	8.0	6.49
50.1	63.6	6.25	244.7	8.0	8.71
59.9	35.6	7.05	239.8	8.0	9.43
70.1	13.0	7.66	235.0	8.0	9.84
79.9	10.2	8.02	242.5	8.0	10.97

5.3 The enclosed mass

At the Sun's own radius, equation (5) with the adopted Θ_0 gives

$$M(<R_0) \simeq \frac{(236)^2 \times 8.15}{4.30091 \times 10^{-6}} \simeq 1.06 \times 10^{11}\,M_\odot. \tag{8}$$

Our outermost measurement evaluates to $M(<R_0) \approx 1.10 \times 10^{11}\,M_\odot$, which is consistent with the theoretical expectation of equation (8). It is crucial to note that this represents the total dynamical (gravitating) mass required by a spherical approximation, not just a dark-matter mass. The stars, interstellar gas, and the Galactic bulge all contribute significantly to this total; indeed, inside the solar circle, baryonic matter accounts for a substantial fraction of the potential.

6 Discussion

6.1 What the measurement supports

The defensible conclusion from this analysis is that the inner Milky Way possesses an extended gravitating mass distribution. The derived rotation curve plateaus near $235\,\mathrm{km\,s^{-1}}$, exhibiting no signs of the Keplerian decline ($\Theta \propto R^{-1/2}$) expected if the Galaxy's mass were entirely concentrated at its center like a point mass. The tangent-point method inherently stops at $R = R_0$ due to the geometric limitation $R_{\mathrm{tan}} = R_0|\sin l| \leq R_0$, rather than a lack of hydrogen gas. Because this accessible region ($R \leq R_0$) is precisely where visible baryonic matter (the disk and bulge) contributes most strongly to the gravitational potential, we cannot definitively map an extended dark matter halo. A complete dark-matter mass model requires baryonic mass subtraction and kinematic tracers extending well beyond the solar circle [1].

6.2 Systematics

Ultimately, the primary difficulty in this experiment lies in the systematic uncertainties of the disk's kinematics, not instrumental sensitivity. The HI4PI survey's 0.043 K channel noise sits more

Table 3: Assumptions and the errors they produce.

Assumption or issue	Why it matters, and what it did here
Circular gas orbits	The bar and the spiral arms drive non-circular motions of several to $\sim 10\,\mathrm{km\,s^{-1}}$; the innermost points are the least reliable. Measurements at $l < 20^\circ$ ($R \lesssim 2.8\,\mathrm{kpc}$) remain marginally affected by the bar potential.
Edge definition	Turbulence and beam smearing broaden the edge, and the threshold choice can move v_{term} by $\sim 10\,\mathrm{km\,s^{-1}}$. Usually the largest single systematic. Empirically calibrated to be an offset of $13.6\,\mathrm{km\,s^{-1}}$.
Velocity frame	The radio and dynamical LSR differ by up to $\sim 3\,\mathrm{km\,s^{-1}}$, mattering most as $l \to 90^\circ$ [5]. Implicitly absorbed and corrected by our $13.6\,\mathrm{km\,s^{-1}}$high-longitude calibration.
Adopted R_0, Θ_0	$\pm 0.15\,\mathrm{kpc}$ and $\pm 7\,\mathrm{km\,s^{-1}}$ propagate into every point of the curve. Contributes $\approx 8\,\mathrm{km\,s^{-1}}$to $\delta\Theta$ and $\approx 7\%$ to the mass uncertainty.
Spherical mass formula	The disk is flattened, so $\Theta^2 R/G$ overestimates the true enclosed mass and is a dynamical scale rather than a precision measurement.

than two orders of magnitude below our 10.0 K edge criterion. Therefore, effort spent minimizing receiver noise is effort spent on the wrong problem; defining the physical edge of turbulent gas remains the dominant limitation.

7 Conclusions

1. The circular-speed curve of the inner Milky Way was measured from $R \approx 2.1\,\mathrm{kpc}$ to $R = 8.15\,\mathrm{kpc}$ using 21 cm H I emission spectra from the HI4PI survey.
2. The rotation curve rises steeply in the inner Galaxy before flattening into a sustained plateau near $235\,\mathrm{km\,s^{-1}}$.
3. The enclosed dynamical mass within the solar circle is calculated as $M(< R_0) = (1.10 \pm 0.08) \times 10^{11} M_\odot$.
4. The definition of the turbulent gas edge is the dominant systematic uncertainty, which a high-longitude geometric test empirically calibrated to a $13.6\,\mathrm{km\,s^{-1}}$ offset.
5. While constraining the total interior dynamical mass, these kinematics alone cannot differentiate between baryonic and dark matter contributions, nor can they map the halo beyond the solar circle.

Acknowledgements

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A Propagation of uncertainties

From equation (4), with $v_{\rm term}$ and Θ_0 independent,

$$\delta\Theta = \sqrt{(\delta v_{\rm term})^2 + (|\sin l|\,\delta\Theta_0)^2}, \qquad \delta R = |\sin l|\,\delta R_0, \tag{9}$$

and from equation (5), since $M \propto \Theta^2 R$,

$$\frac{\delta M}{M} = \sqrt{\left(\frac{2\,\delta\Theta}{\Theta}\right)^2 + \left(\frac{\delta R}{R}\right)^2}. \tag{10}$$

This error propagation is strictly conservative because R_0 and Θ_0 are not independent variables; their ratio $\Omega_0 = \Theta_0/R_0$ is constrained much more tightly by proper motion measurements than either parameter is individually [4]. Consequently, adding their isolated errors in quadrature artificially inflates the true physical uncertainty of the derived mass.

B Formula sheet

$$\Delta E = h\nu_0, \qquad \nu_0 = 1420.405752\ \text{MHz}, \qquad \lambda_0 = c/\nu_0 = 21.106\ \text{cm};$$

$$T_B = \frac{c^2}{2k_B\nu^2} I_\nu, \qquad v_{\text{rad}} \simeq c\,\frac{\nu_0 - \nu_{\text{obs}}}{\nu_0};$$

$$v_{\text{LSR}} = \left[\Theta(R)\frac{R_0}{R} - \Theta_0\right]\sin l = R_0\left[\Omega(R) - \Omega_0\right]\sin l;$$

$$d_{\text{tan}} = R_0\cos l, \qquad R_{\text{tan}} = R_0|\sin l|, \qquad \Theta(R) = |v_{\text{term}}| + \Theta_0|\sin l|;$$

$$v_{\text{measured}} - v_{\text{term}} = \sqrt{2}\,\sigma\,\text{erfc}^{-1}\left(\frac{2T_{\text{thr}}}{T_{\text{edge}}}\right), \qquad v_{\text{term}}^{\text{ideal}} = 2AR_0(1-\sin l);$$

$$M(<R) = \frac{\Theta^2(R)R}{G}, \qquad \Theta \approx \text{const} \Rightarrow M(<R) \propto R, \qquad \rho \propto R^{-2}.$$